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# Does Social Bot Help Socialize? Evidence from a Microblogging Platform

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**Abstract.** Leveraging advancements in large language models, social media platforms are increasingly deploying sophisticated chatbots, termed social bots, with the potential to stimulate user interaction. However, concerns linger regarding the socializing value of these bots in *public* settings. We investigate this phenomenon using data from the launch of CommentRobot on a microblogging platform. Analyzing user interactions with this platform-owned bot, we find that posts receiving bot-generated comments experience increased user engagement, demonstrating the socializing value of social bots at the post level. Results from an online experiment confirm this finding and reveal that the socializing value stems from both bot identity and high-quality content. Mechanism tests suggest that the quality of bot-generated comments—particularly their attractiveness, relevance, and inclusion of social cues—significantly influences user engagement. Moreover, we evaluate existing bot targeting strategies and propose policy learning-based improvements to optimize engagement. Despite the positive impact on post-level engagement, we find that receiving bot comments primarily encourages future bot-related posts rather than increasing overall user posting activity, contrary to platform expectations. Theoretically, this study contributes to the literature on social bots and the "Computers are Social Actors" framework by empirically examining relevant constructs in a novel context. Practically, our findings highlight the need for platforms to refine social bot deployment strategies to maximize user engagement while mitigating unintended consequences.

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## 1. Introduction

With signs of user fatigue and disengagement raising concerns about the future of social media, especially for established social networking sites such as X (formerly Twitter) and Facebook (Faverio and Sidoti 2024),[1] platforms are turning to advancements in artificial intelligence (AI) as a potential solution to counteract this trend. One approach involves providing AI tools to aid user content generation (e.g., Zhang et al. 2024), whereas another leverages the capabilities of large language models (LLMs) to develop sophisticated chatbots (henceforth referred to as *social bots*). Indeed, recent breakthroughs in LLMs have enabled social bots to provide high-quality, relevant content and mimic complex human behaviors, including cognition (Dhingra et al. 2023) and empathy (Elyoseph et al. 2023), which create opportunities for boosting user engagement on a large scale. As a result, social media platforms are introducing social bots that go beyond their traditional roles for operational tasks (e.g., content moderation). For instance, premium users on X can engage in private conversations with a chatbot named Grok (Agius 2023). In such *private* settings, although previous studies have shown the socializing value of chatbots (e.g., showing empathy) (Ta et al. 2020), concerns persist regarding the negative consequences stemming from bot misbehavior. For example, a parcel delivery firm disabled its chatbot after the bot was tricked into writing a poem about bad customer service provided by the company (Reuters 2024).

In *public* settings like social media platforms, the stakes of deploying social bots could be even higher. Unlike private interactions, public conversations between users and social bots are visible to a broader audience, increasing the risk of negative publicity from bot misbehavior. Yet, the potential benefits of social bots for increasing engagement and boosting user posting activity on social media platforms are considerable, particularly with the growth of the user base and potential monetization strategies for human-bot interactions. In essence, this double-edged sword feature of social bots raises the following questions: Do social bots help social interactions in public settings? Will social media users appreciate their peers' public interactions with a social bot? Or will they be reluctant to engage with them in a public setting?

On the one hand, according to the "Computers are Social Actors" (CASA) framework (Gambino et al. 2020), social media users may be impressed by the high quality of bot-generated content (e.g., social cues) and thus consider the bot a good social actor. With such an impression, users may be more willing to engage with their peers' public interaction with a social bot. On the other hand, because of the ample anecdotal evidence of potential low-quality bot-generated content (e.g., hallucinations), social media users may be algorithm-averse (Liu et al. 2023) and exhibit less inclination to engage with the conversation between their peers and a social bot.

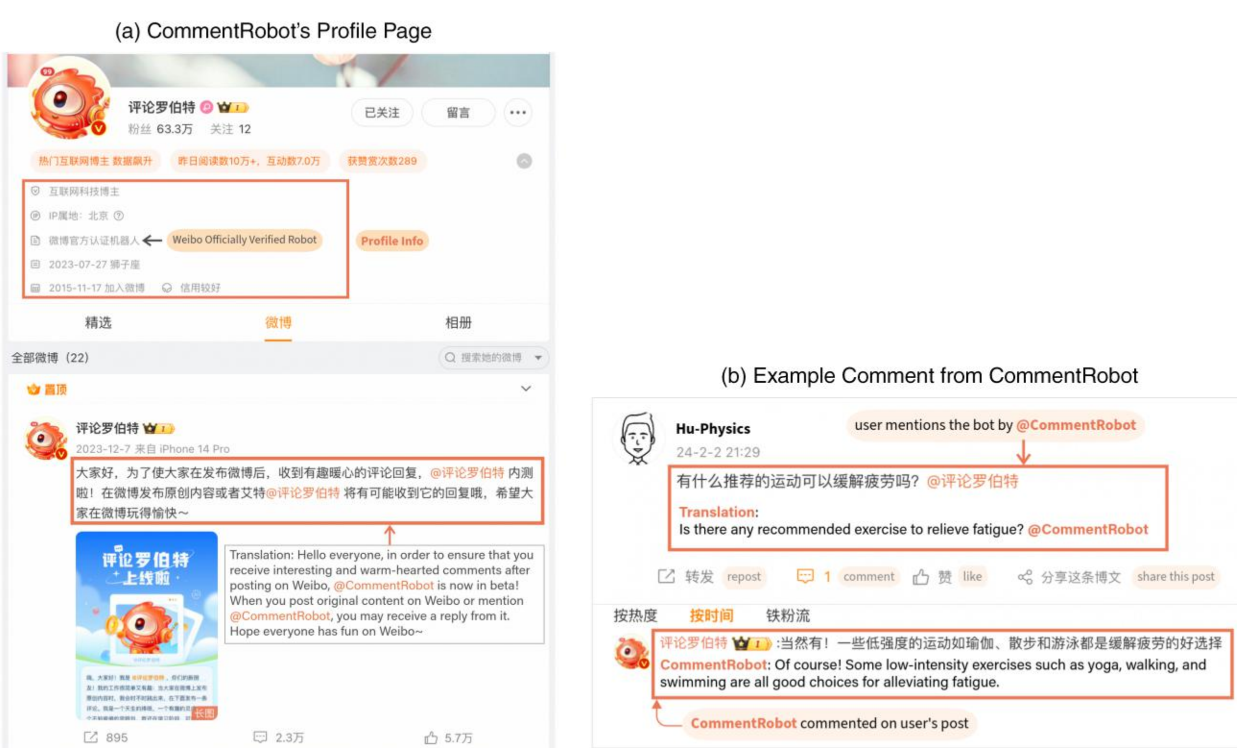
Despite the crucial practical importance and the mixed theoretical predictions, the previous literature on conversational AI and social bots (see a summary in Online Appendix Table A.1) provides no empirical evidence on the socializing value of social bots in public human-bot interactions. Indeed, prior studies have predominantly focused on private human-AI interactions and the functional roles of social bots, leaving their socializing value in public settings largely unexplored. As such, this study aims to fill this research gap by addressing the following research questions: *Do social bots hold socializing value in public human-bot interactions? If so, how is this value delivered, and what are its boundaries?*

To this end, we leverage the recent launch of an LLM-based bot, CommentRobot, on Weibo, a leading microblogging platform in China. This platform-owned social bot is designed to automatically comment on user posts, functioning as a social actor (see Figure 1 for illustration). Its beta launch was in December 2023, aiming at enhancing user experience and stimulating posting activity. We collected all posts explicitly mentioning @CommentRobot in January 2024, after its official launch, to analyze the audience's reactions to their peers' interactions with the bot.

We conduct the main analysis at the post level, focusing on posts representing the author's first interaction with the bot. Our results indicate that receiving bot-generated comments leads to more likes and comments from peer users on a post, thereby supporting the socializing value of social bots at the post level for the focal user. To address concerns regarding the nonrandom assignment of bot comments, we conduct an instrumental variable analysis, and the main findings remain consistent across a battery of robustness checks and are further supported in an online randomized experiment.

Moreover, we delve into mechanisms underlying the observed positive socializing value. Results of the online experiment indicate that both bot identity and comment content contribute to increased user engagement. Consistent with the CASA framework, qualitative analysis

**Figure 1.** (Color online) *CommentRobot* on Weibo



of participant responses to open-ended questions in the experiment highlights the critical role of bot comment quality in shaping user interactions. Findings from observational data reinforce this conclusion, showing that higher-quality bot comments—those that are more attractive and relevant, and have more social cues—elicit greater user engagement than comments with lower values on these attributes.

With rich post- and user-level characteristics, we evaluate the efficacy of the existing targeting strategy employed by the bot when generating comments. The findings indicate a preference for responding to users who do not mention (i.e., @) others in their posts, which is deemed effective for fostering peer users' engagement. However, our findings also uncover several strategies that fail to optimize user engagement. For example, although the bot prefers to comment on less active users, our results suggest that active users benefit more from receiving a bot comment than idle users. To improve bot targeting efficiency, we adopt a policy tree approach (Athey and Wager 2021) to identify assignment strategies that maximize user engagement. Our findings demonstrate that leveraging treatment heterogeneity significantly improves targeting efficiency. The optimized policies consistently yield higher engagement effects than the platform's existing strategy, even when only post-level or user-level characteristics are used for targeting.

Besides boosting engagement at the post level, platforms also aim to increase the posting activity of human users. To investigate whether human-bot interactions boost future posting activities from focal users, we exploit the stacked difference-in-differences (DID) design (Cengiz et al. 2019, Zhang et al. 2024). We find significantly positive effects on future bot-related posts but insignificant effects on future original posts and reposts for users who receive bot comments compared with those who do not. This result implies that the observed socializing value at the post level extends to increased posting behavior among focal users on the platform, but is only limited to those bot-related posts, which fails to meet platform expectations of stimulating overall posting activity.

The rest of the paper is organized as follows. We review the relevant literature in Section 2 and provide the theoretical development in Section 3. In Section 4, we describe the research context, data collection, and measurement of key variables. In Sections 5 and 6, we discuss identification issues and report the results of various empirical tests. We conclude the paper in Section 7 by discussing the implications and limitations of this study.

## 2. Literature Review

Our study is broadly related to the literature on conversational AI and specifically related to the literature on social bots. We review these two streams of literature separately in this section. Table A.1 in the Online Appendix summarizes the relevant studies in the Information Systems (IS) literature.

### 2.1. Conversational AI

The IS literature on conversational AI primarily focuses on chatbot attributes and their impacts on users. In particular, a significant portion of this stream examines the effects of anthropomorphism and human-like attributes on user perceptions and behaviors. For instance, Schuetzler et al. (2020) investigated how the conversational skills of a chatbot influence perceived social presence and anthropomorphism, which ultimately affects user perception. Expanding on this, Seeger et al. (2021) developed a framework for designing anthropomorphic chatbots, and found that a mix of verbal and nonverbal cues significantly enhances perceived anthropomorphism. In terms of human-like attributes, Schanke et al. (2021) explored the influence of attributes such as communication delays and social presence in customer service chatbots on transaction conversion. Furthermore, Chandra et al. (2022) investigated how cognitive, relational, and emotional competencies in conversational AI agents affect user engagement.

Although anthropomorphism and human-like attributes remain the central theme, recent studies have begun to explore additional crucial attributes of conversational AI. Gnewuch et al. (2024) examined hybrid service agents that integrate AI and human agents, focusing on the effects of disclosing human involvement in these systems on user interactions. The emotional dynamics of AI interactions have also garnered attention. Han et al. (2023) demonstrated that positive emotions expressed by AI agents can significantly influence customer service evaluations both affectively and cognitively. The deployment context of conversational AI plays an essential role as well. Brendel et al. (2023) studied user responses to errors made by human-like and non-human-like conversational agents, noting that perceived humanness can affect user aggression. In a distinct context, Sachdeva et al. (2024) investigated the use of chatbots for collecting customer reviews, illustrating that chatbots can affect users' thought processes and adherence to genre norms during review writing.

### 2.2. Social Bots

Social bots are AI-powered automated programs specifically designed to operate on social media. Prior literature on social bots has predominantly focused on their pivotal role in information dissemination and methods for their detection. For example, Shao et al. (2018) conducted an extensive analysis of Twitter data, revealing that social bots disproportionately amplify content from low-credibility sources, particularly during the early stages of information dissemination. Furthermore, a following study by Salge et al. (2022) developed a novel framework of algorithmic conduit brokerage to understand how bots perform information dissemination and the design choices that may influence their actions. Amidst concerns regarding misinformation, researchers proposed machine learning-based methods to detect bots on social media (Orabi et al. 2020). Benjamin and Raghu (2023) further developed a framework to augment deep learning models by leveraging human reactions to bot messages to enhance social bot detection.

With the prevalence of social bots, recent research has started to explore their impact on human activities. Delkhosh et al. (2023) revealed a positive correlation between bots' voting behavior and human user engagement in blockchain-based social media. Concurrently, platforms have deployed their social bots, prompting researchers to investigate their capacity to stimulate human user activities. He et al. (2024) explored the influence of platform-owned bot moderators on human moderators' community-policing efforts, demonstrating that the introduction of bot moderators can encourage human moderators to moderate more posts. Safadi et al. (2024) investigated how different types of bots (reflexive versus supervisory) influence human-to-human interaction patterns in online communities.

Our study contributes to the literature on conversational AI and social bots by examining the socializing value of platform-owned social bots in public settings, distinguishing itself from previous studies. First, unlike prior studies that mainly focus on one-to-one, private AI-human interactions, our research delves into the dynamics of public human-bot interactions observable to all users on the platform. Second, this study centers on a social bot specifically designed for social interactions with users, as opposed to performing purely functional tasks. This contrasts with studies such as He et al. (2024) and Safadi et al. (2024), where bots were primarily engaged in automating moderation or summarizing text, adhering to fixed rules and producing predictable outcomes. Our study explores a bot endowed with generative capabilities, a distinct social identity, and the ability to engage in nuanced, context-dependent interactions in response to social requests from users.

## 3. Theoretical Development

With social bots' increasing popularity, the CASA framework provides a vital theoretical lens to understand users' interactions with these technologies. CASA posits that humans apply social scripts from human-human interactions when engaging with computers (Nass and Moon 2000). This framework has recently been extended to human interactions with modern technological artifacts, such as voice assistants and social bots (Gambino et al. 2020).

Historically, CASA research has primarily focused on dyadic human-bot interactions, investigating how bot characteristics influence immediate outcomes within the same interaction. However, social media platforms offer new affordances for interactions with social bots (Karahanna et al. 2018). The public nature of these platforms implies that human-bot interactions are visible to all users, transforming what would typically be private, two-sided interactions into multisided public ones. This visibility allows peer users to observe and participate in these interactions through engagement mechanisms (e.g., likes and comments). Therefore, drawing upon the CASA framework, we formulate potential directions for the effects of public human-bot interactions on peer users' engagement.

On the one hand, a social bot might function as a *good* social actor when it generates high-quality content, encouraging peer user engagement. This effect may stem from social proof, where people look to others' behavior to guide their actions (Cialdini et al. 1990). Traditionally studied in human-to-human interactions (e.g., Hilverda et al. 2018), social proof can extend to human-AI interactions if users perceive bots as good social actors rather than automated systems. When bots achieve this perception, users are more likely to interpret bot interactions similarly to human interactions, activating social proof mechanisms.

Whether a bot can act as a good social actor depends on its functional competence and social intelligence (Chaves and Gerosa 2021). A bot's functional competence, defined as its ability to process and respond to human requests effectively, is reflected by the *relevance* of bot-generated content to user requests. Modern bots, often powered by advanced LLMs, excel at generating contextually relevant responses (Yang et al. 2024). These tailored responses signal the bot's cognitive capabilities in understanding user requests and adapting to different conversational contexts, which is usually appreciated by the audience and thereby boosts user engagement (Feine et al. 2019).

Social intelligence, representing a more advanced functionality of bots, entails their ability to engage in social interactions in human-like ways (Dautenhahn 2007). Demonstrating human-like *social cues* is crucial for social bots to function effectively as social actors (Mathur et al. 2024), because these social cues help attribute human traits to nonhuman entities (Schanke et al. 2021), aligning with the notion of anthropomorphism where conversational AI enables human-like interactions between chatbots and users (Han et al. 2023). LLM-based bots have shown substantial capability in delivering various social cues, with studies demonstrating that bots mimicking human language can embed social cues in generated text, enhancing user engagement (Elyoseph et al. 2023).

The public nature of social media introduces a new construct through which social bots can deliver their socializing value, the *attractiveness* of the bot-generated content to the audience. Unlike private interactions, public human-bot interactions on social media platforms allow the audience to directly observe and evaluate the engaging nature of bot-generated content through visible metrics such as likes. The bot's capability to engage and attract the audience is particularly crucial on social media platforms where captivating content is key to retaining user attention (Radziwill and Benton 2017). Empowered by advanced LLMs, social bots could generate attractive content, increasing social interaction and stimulating greater user engagement (Li et al. 2021).

On the other hand, an LLM-based social bot that generates low-quality content might function as a *bad* social actor. A low relevance of responses to the request usually signals the bot's technical inability and may even produce hallucinations and misinformation (McIntosh et al. 2023), thereby discouraging peers from engaging with the public human-bot interactions (Hertzum and Hornbæk 2023). As for social cues, social bots may learn bad human traits from the training data. For example, Si et al. (2022) show that publicly available bots are susceptible to providing toxic responses when exposed to toxic queries; Beattie et al. (2022) find that bots may inadvertently perpetuate biases learned from training data, including gender and racial biases, in their interactions with users. This concern is particularly pertinent in the context of social media, where training data often include negative and even offensive human interactions (Zampieri et al. 2019). Consequently, there is a risk of algorithm aversion developing among users if bots exhibit improper social behavior, potentially resulting in decreased engagement (Liu et al. 2023).

In summary, the socializing value of a social bot crucially depends on its role as a social actor shaped by its generated content. If a bot acts as a good social actor, we anticipate a positive socializing value. If a bot acts as a bad social actor, human-bot interactions may harm user engagement. Motivated by these mixed theoretical predictions, we employ empirical tests and mechanism analyses to understand the effects of public human-bot interactions on peer users' engagement.

# 4. Data

## 4.1. Research Context

Our data source is Weibo, the leading microblogging platform in China, with over 260 million daily active users as of 2024 (Statista 2024). This study focuses on CommentRobot, a social bot owned by the platform and powered by LLMs (see Online Appendix Figure G.1). The official account page of CommentRobot on Weibo, displayed in Figure 1(a), clearly indicates its bot identity through the "*Weibo Officially Verified Robot*" tag in its profile information. The bot's inaugural post was made on July 27, 2023, providing no specific information. On November 7, 2023, the bot encouraged users to mention it (i.e., include @CommentRobot) in their posts, hinting at the possibility of receiving bot comments (see the bottom left post in Online Appendix Figure G.2). However, the specifics about this bot were only disclosed on December 7, 2023, when CommentRobot announced its beta launch and provided guidelines for expediting comment reception (see the bottom left post in Figure 1). In the image of a post (see Online Appendix Figure G.1), CommentRobot highlighted the platform's objective in deploying this social bot: to enhance user experience and stimulate posting activity.

The primary function of CommentRobot is to automatically generate comments on users' original posts or on posts that mention (@) the bot, as illustrated by the example in Figure 1(b). Unlike regular users, the bot does not follow other users, and its capabilities are limited to comment generation, without engaging in sharing, liking, or other interactions. According to the announcement, a proactive approach for invoking the bot's comment-generating function is to include @CommentRobot in their post. However, because of limited capacity, the bot cannot leave comments on all posts mentioning it. Approximately 60% of posts in our sample did not receive bot comments.

## 4.2. Data Collection

Although posts without mentions of the bot may still receive bot comments, we focus on posts with @CommentRobot mentions for two reasons. First, from a technical standpoint, Weibo's data structure is hierarchical, meaning that a post must first be collected to retrieve all of its comments, which are subsequently used to determine if bot comments are received. Thus, @CommentRobot, serving as the keyword, becomes necessary to identify bot-related posts and retrieve their comments. Second, restricting the sample to posts with an explicit mention of the bot helps identify the causal effect. Users' motivations underlying their posting behavior are difficult to observe directly. However, explicit bot mentions could signal a strong willingness to interact with the social bot, suggesting that these users are comparable in terms of underlying motivations.

We started the data collection process by searching for all posts containing @CommentRobot published in January 2024. We chose this time frame to ensure that collected interactions occurred after the bot's official launch. Reposts were excluded from the data set to retain only original posts. This process led to a total of 106,523 posts authored by 64,365 users. Additionally, we collected users' online profile information and their historical posts published in January 2024. In the main analysis, to identify the audience's reactions to the first user-bot interaction, we focus on 64,365 posts in which the user mentions @CommentRobot for the first time.

### 4.3. Variables

Table 1 provides the definitions and summary statistics for key variables used in the empirical analysis. The outcome variable measures user engagement with the post, quantified by the number of likes and comments. Because of high skewness, we use the log-transformed measures as the outcome variables. Bot-generated comments and those from the post author or before a bot comment are excluded when constructing the outcome variable *LogComments*. The key independent variable is *BotComment*, which equals one if the bot comments on the post and zero otherwise.

The nonrandom assignment of bot comments is a major endogeneity concern. Specifically, the bot might selectively target certain types of posts or users, introducing potential selection bias. To alleviate this concern, we incorporate a variety of user- and post-level characteristics as control variables. User characteristics are derived from online profiles. Specifically, *LogFollowers* and *LogFollowings* represent the log-transformed number of followers and followings, respectively. *LogUpdates* is the log-transformed number of posts published by the user. *Verified* is a binary indicator of the user account's verification status. *Male* equals one if the user identifies as male on Weibo. *PaidUser* indicates whether the user is paying for premium content access. *UserTenure* quantifies the number of years since the user's registration.

To account for post characteristics that might impact user engagement, we construct the following variables for each post. *ContentLength* represents the word count of the post. *NumPic* is the number of pictures included in the post, whereas *Video* is a binary variable indicating the presence of a video, both reflecting the visual appeal of the content. *Hashtag* is a binary variable denoting the use of hashtags. *NumEmoji* captures the number of emojis in the post, reflecting its emotional tone. Additionally, *AtOthers* indicates whether a post mentions other users, and *AtBotAsStart* is a binary indicator of whether the post starts by mentioning the bot. Moreover, we conduct latent semantic analysis (LSA) to capture the semantic meanings of posts, the details of which are reported in Online Appendix C.

## 5. Empirical Analysis

To examine the effect of public human-bot interactions on user engagement, we estimate the following ordinary least squares (OLS) model with time fixed effects at the post level, indexed by $i$:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 Z_i + \delta_t + \epsilon_{i,t}. \qquad (1)$$

The outcome variable $Y_i$ is the engagement metric for post $i$, which is measured by either *LogLikes* or *LogComments*. The key coefficient of interest, $\beta_1$, captures the main treatment effect of receiving bot-generated comments on user engagement. $X_i$ includes the author's characteristics, such as their online profile. $Z_i$ includes the characteristics of post $i$, such as the number of pictures and content clusters. $\delta_t$ refers to time fixed effects including the date and the hour-of-the-day fixed effects.

### 5.1. Main Results

Table 2 reports the estimation results of Model (1). We gradually add controls and fixed effects to demonstrate

**Table 1.** Summary Statistics

| Variable | Mean | Std. Dev. | Definition |
|---|---|---|---|
| Outcome measures | | | |
| *LogLikes* | 0.49 | 0.78 | Log-transformed number of likes a post receives |
| *LogComments* | 0.71 | 0.78 | Log-transformed number of comments a post receives |
| Treatment | | | |
| *BotComment* | 0.40 | 0.49 | Binary variable indicating whether a bot comments on a post |
| Post characteristics | | | |
| *ContentLength* | 30.38 | 63.06 | Number of words in a post |
| *NumPic* | 0.17 | 0.85 | Number of pictures in a post |
| *Video* | 0.001 | 0.04 | Binary variable indicating whether a post contains videos |
| *Hashtag* | 0.06 | 0.36 | Binary variable indicating whether a post includes hashtags |
| *NumEmoji* | 0.49 | 2.06 | Number of emojis used in a post |
| *AtOthers* | 0.06 | 0.45 | Binary variable indicating whether a post mentions (@) other users |
| *AtBotAsStart* | 0.57 | 0.49 | Binary variable indicating whether a post mentions (@) CommentRobot at the beginning |
| User characteristics | | | |
| *LogFollowers* | 4.80 | 2.22 | Log-transformed number of followers the user has |
| *LogFollowings* | 5.80 | 1.04 | Log-transformed number of accounts the user is following |
| *LogUpdates* | 7.19 | 1.60 | Log-transformed number of posts ever published by the user |
| *Verified* | 0.24 | 0.42 | Binary variable indicating whether a user account is verified |
| *Male* | 0.17 | 0.37 | Binary variable indicating whether a user is self-identified as male |
| *PaidUser* | 0.01 | 0.11 | Binary variable indicating whether a user pays for premium content access |
| *UserTenure* | 5.34 | 3.22 | Number of years since a user's registration |

*Notes.* The number of observations is 64,365. LSA variables are not reported in this table.

**Table 2.** Main Analysis

| | *LogLikes* | | | *LogComments* | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *BotComment* | 0.0951*** | 0.1151*** | 0.1087*** | 0.2389*** | 0.2378*** | 0.2332*** |
| | (0.0064) | (0.0063) | (0.0062) | (0.0060) | (0.0058) | (0.0058) |
| User controls | No | Yes | Yes | No | Yes | Yes |
| Post controls | No | Yes | Yes | No | Yes | Yes |
| Date FEs | No | No | Yes | No | No | Yes |
| Hour-of-day FEs | No | No | Yes | No | No | Yes |
| Observations | 64,365 | 64,365 | 64,365 | 64,365 | 64,365 | 64,365 |
| $R^2$ | 0.004 | 0.164 | 0.169 | 0.023 | 0.183 | 0.189 |

*Notes*. Robust standard errors are in parentheses. FEs, fixed effects.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

the robustness of the findings. Specifically, columns (1) and (4) report the results without controls and fixed effects, columns (2) and (5) report the results with controls but without fixed effects, and columns (3) and (6) report the results incorporating both controls and fixed effects. Columns (1)–(3) measure user engagement with *LogLikes*. Across different specifications, the coefficients of *BotComment* are consistently positive and statistically significant, indicating that the presence of bot-generated comments leads to an increase in the number of likes garnered by a post. In terms of the magnitude, column (3) suggests that receiving a bot comment leads to approximately an 11% increase in the number of likes received. As evident by columns (4)–(6), the coefficients of *BotComment* are significantly positive when *LogComments* is the outcome variable, suggesting that bot-generated comments attract more comments from peer users. In terms of the magnitude, the presence of a bot comment leads to a 23% increase in the number of comments. Overall, these findings highlight the positive impact of bot-generated comments on user engagement, thereby demonstrating the socializing value of social bots for the focal author in public settings.

## 5.2. Instrumental Variable Analysis

Despite including a rich set of controls and fixed effects in Model (1), concerns may still arise regarding an omitted variable(s) that could influence the likelihood of receiving a bot comment and potentially correlate with user engagement levels. To alleviate this concern, we leverage an IV to identify the causal effect of bot-generated comments on user engagement.

As shown in Figure G.2 in the Online Appendix, the platform-owned social bot, CommentRobot, published several posts on Weibo to promote its presence and encourage user interactions. We collected all posts from CommentRobot before our observation window (i.e., January 2024) and checked if users in our sample had commented on these posts. Next, we constructed a binary variable, *BotInteractionBefore*, which equals one if a user commented on bot posts before January 2024 and zero otherwise. To serve as a valid IV, *BotInteractionBefore* must satisfy two restrictions: *relevance restriction*, which requires a strong correlation between the IV and the endogenous variable (i.e., *BotComment*), and *exclusion restriction*, which requires the IV to affect the outcome variables *only* through the endogenous variable.

For the relevance restriction, although the bot's targeting strategy (e.g., criteria for selecting users and posts) remains opaque, we expect that the bot needs a solid starting point, and users who previously interacted with its posts may serve as such. In other words, users with prior interactions with the bot might be more likely to receive bot comments. Following the econometrics literature, we exploit the F-statistics of the first-stage analysis to formally test whether the relevance restriction is satisfied. For the exclusion restriction, we argue that commenting on bot posts before January 2024 would not directly affect the engagement level of a post published in January. The reasons are twofold. First, Weibo does not publicly display a user's interaction with bot posts on their timeline, making it rather difficult for others to be aware of such activities.[2] Second, a concern could be that commenting on bot posts might attract attention from other users interested in the bot, who may later engage with the user's bot-related posts. To alleviate this concern, we examined the number of likes received by comments under bot posts before January 2024, with over 95% of these comments receiving zero likes, indicating the absence of an attention-drawing mechanism.

Table 3 presents the results of the two-stage least squares analysis (2SLS), with *BotInteractionBefore* serving as the IV. Column (1) reports the results of the first-stage analysis, where the coefficient of *BotInteractionBefore* is significantly positive, indicating that users with prior interactions with the bot are more likely to receive a bot comment. More importantly, the F-statistic is 17.15, which is above the critical value of 10 suggested by Stock and Yogo (2005), indicating that the relevance restriction is satisfied and the IV is not weak. Columns (2) and (3) report significantly positive coefficients of

**Table 3.** Instrumental Variable Analysis

| | First stage | Second stage | |
|---|---|---|---|
| | *BotComment* (1) | *LogLikes* (2) | *LogComments* (3) |
| *BotInteractionBefore* | 0.0369*** (0.0089) | | |
| *BotComment* | | 1.5255*** (0.5275) | 3.0064*** (0.7809) |
| User controls | Yes | Yes | Yes |
| Post controls | Yes | Yes | Yes |
| Date FEs | Yes | Yes | Yes |
| Hour-of-day FEs | Yes | Yes | Yes |
| F-stat | 17.15 | | |
| Observations | 64,365 | 64,365 | 64,365 |

*Note.* Robust standard errors are in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

*BotComment* when engagement metrics are the outcome variables, demonstrating the robustness of our main findings.

### 5.3. Robustness Checks

We conduct a series of tests to ensure the robustness of our findings. Table 4 summarizes these checks, with details available in Online Appendix B. To address concerns about reverse causality, we reestimate our model using only posts where the bot comment appears first, ensuring that peer engagement follows the bot interaction instead of the bot targeting highly engaged posts to comment; we conduct a subsample analysis focusing on posts where bot comments occurred extremely fast following the post, ensuring that human likes will very likely follow the bot comment. The results, reported in Table B.1 and Table B.2 in the Online Appendix, remain consistent with our main findings.

To assess robustness to alternative empirical specifications, we implement several additional tests. First, we introduce user fixed effects to account for unobserved heterogeneity (Online Appendix Table B.3). Second, we control for post categories (see definitions and measurements in Online Appendix E) to ensure that engagement effects are not driven by differences in content topics (Online Appendix Table B.4). Third, we estimate two commonly used count models, Poisson and Negative Binomial (Online Appendix Table B.5). Finally, we apply coarsened exact matching (CEM) to improve comparability between posts with and without bot comments, with results shown in Table B.6 in the Online Appendix.

To address concerns about the exogeneity of *BotInteractionBefore*, we examine whether users with prior bot interactions are more likely to make AI-related posts. Using a list of AI-related keywords, we find that such posts are rare and that posting behavior and engagement levels do not significantly differ between users with and without prior bot interactions (Online Appendix Table B.7), reinforcing the validity of *BotInteractionBefore* as an instrumental variable.

**Table 4.** Summary of Robustness Checks

| Robustness check | Description | Online appendix table(s) |
|---|---|---|
| Reverse causality | Reestimates model using only posts where the bot comment appears first to ensure engagement follows bot interaction. | Table B.1 |
| | A subsample analysis focusing on posts where bot comments appeared very shortly after the original post. | Table B.2 |
| Alternative specifications | Introduces user fixed effects to control for unobserved heterogeneity. | Table B.3 |
| | Includes post category controls to ensure engagement effects are not driven by content differences. | Table B.4 |
| | Estimates Poisson and Negative Binomial models to account for nonnormality in engagement metrics. | Table B.5 |
| | Applies coarsened exact matching to improve comparability between posts with and without bot comments. | Table B.6 |
| Instrumental variables | Checks if users with prior bot interactions are more likely to post about AI. Finds no significant difference in posting behavior or engagement. | Table B.7 |
| | Uses bot mention traffic before and after posts as an alternative IV. | Table B.8 |
| | Controls for influencer activity to ensure engagement effects are not driven by high-profile users attracting disproportionate attention. | Table B.9 |
| | Restricts sample to high-traffic hours to test IV stability across varying posting conditions. | Tables B.10 and B.11 |
| | Applies a double machine learning framework using partially linear IV regression models for flexible control variable relationships. | Table B.12 |
| | Estimates a model incorporating both IVs and performs an overidentification test to validate the instruments. | Table B.13 |
| Long-term effect | Reestimates model using February-March 2024 data to rule out effects driven by initial LLM hype. | Table B.14 |
| Posts without @CommentRobot | Analyzes posts mentioning “CommentRobot” without @ to test if unfulfilled bot requests drive engagement. | Table B.15 |

We also introduce *BotTraffic* as an alternative IV, measuring the number of bot-mentioning posts published immediately before and after a focal post. Because the bot's commenting capacity is limited, an increase in bot mention requests reduces the likelihood of receiving a bot comment, satisfying the relevance condition. The exclusion restriction is supported by the short time window and the platform's large user base, minimizing the likelihood that these posts directly influence engagement. The 2SLS results in Table B.8 in the Online Appendix confirm the robustness of our findings. Additional robustness tests include controlling for influencer activity to alleviate concerns regarding external events (Online Appendix Table B.9) and restricting the sample to high-traffic hours, both of which confirm the robustness of our IV estimates (Online Appendix Tables B.10 and B.11).

Moreover, we apply a double machine learning framework using partially linear IV regression models, allowing for a flexible, nonlinear relationship between control variables and engagement outcomes. The results, reported in Table B.12 in the Online Appendix, align with the 2SLS estimates. Finally, we estimate a model incorporating both IVs simultaneously and conduct an overidentification test, which supports the validity of the instruments (Online Appendix Table B.13).

To rule out the possibility that our findings are driven by temporary LLM-related hype, we reestimate our model using data from February 1 to March 13, a period before the platform introduced an interactive feature allowing CommentRobot to respond further to post authors' replies. The results, presented in Table B.14 in the Online Appendix, show that *BotComment* remains significantly positive across all specifications, confirming that the observed engagement effects are not a transient phenomenon.

Finally, we test whether our findings might be influenced by unfulfilled bot interaction requests. To do so, we analyze posts where users mention "CommentRobot" without the @ symbol, indicating general discussions about the bot rather than direct requests. The results in Table B.15 in the Online Appendix suggest that unfulfilled requests do not drive our findings, further reinforcing the robustness of our conclusions.

## 5.4. Mechanism Tests

This subsection focuses on validating the mechanism discussed in Section 3. First, we conduct an online experiment to strengthen the internal validity of our main findings and disentangle the sources of the observed effects. A qualitative analysis of participant responses further highlights the role of relevance and social cues in driving user engagement. Second, we use observational data to empirically assess the proposed mechanisms—relevance, social cues, and attractiveness—providing additional support for our theoretical framework.

**5.4.1. Online Experiment.** To enhance the internal validity and disentangle the sources of the effect, we conduct a between-subject randomized experiment on Credamo, an online platform that is widely used by previous research (e.g., Xu et al. 2023) for recruiting Chinese participants. We adopt a factorial design with three experimental conditions: (1) *no-comment*, where posts receive no comment; (2) *human-comment*, where posts are commented on by a human user; and (3) *bot-comment*, where posts receive a comment from the bot. All selected posts are from the January sample of Weibo posts requesting comments from CommentRobot, and the comments shown in the experiment are actual bot-generated responses. Respondents are shown six posts categorized by content: two featuring emotional expressions (one with positive emotion and one with negative emotion), two posing objective questions, and two posing subjective questions. For each post, respondents in human-comment and bot-comment conditions are presented with the same comment. The major difference between these two conditions lies in the commenter's username and profile image. For the human-comment group, we intentionally select usernames and profile images that make it difficult to infer any demographic information, such as gender. For the bot-comment group, the official profile of CommentRobot (i.e., username and profile image) is used to signal the bot identity.

There are two stages in implementing the experiment. In the first stage, we recruit 350 participants, confirming their active usage of Weibo and willingness to participate in our main experiment. Out of these participants, 348 are active Weibo users and agree to participate in the experiment. In the second stage, we randomly assign these participants to three experimental conditions. During the experiment, participants read a paragraph of contextual information (see Online Appendix D) and report their perceptions of CommentRobot. The rationale for doing this is twofold. First, the provided information is what Weibo users can easily access when seeing the post in real life because users can click on the @CommentRobot link in the post to visit the bot's profile page. Second, after presenting the relevant information, we ask respondents to report their perception of the bot identity, which helps evaluate our manipulation and the applicability of the CASA framework.

After that, we display six posts and the corresponding comments (see Online Appendix Figure D.1). After viewing each post and its comment, respondents report their willingness to like and comment using a five-point Likert scale. They are then asked to explain the rationale for their engagement decisions in open-ended responses. We randomize the order of the posts to avoid any order effect. We receive 76 responses in the no-comment group, 62 responses in the human-comment group, and 71 responses in the bot-comment group. To ensure attentiveness, three attention check questions are embedded in the experiment, and all participants pass the checks.

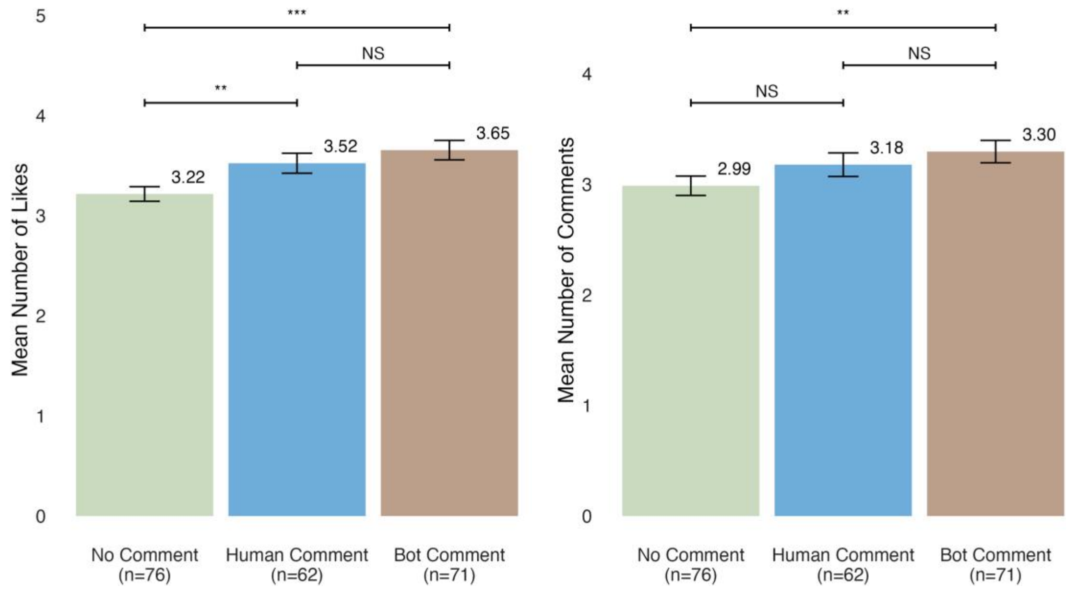
Figure D.2 in the Online Appendix shows that 92.9% of the 209 respondents either agree or strongly agree with their perception of the bot's identity as nonhuman. The balance check results in Table D.1 in the Online Appendix suggest no significant difference between treatment conditions in demographic attributes.

We conduct two-sample *t*-tests to examine the effect of bot-generated comments on user engagement. Figure 2 displays the results. As shown, posts in the bot-comment group receive more likes (diff. = 0.43, $p < 0.01$) and comments (diff. = 0.31, $p < 0.05$) than those in the no-comment group, suggesting that bot-generated comments lead to higher user engagement. Therefore, experiment results concur with those based on observational data, thereby enhancing the internal validity of our findings. Interestingly, Figure 2 helps identify the source of the positive effects. Comparing the human-comment group to the no-comment group, posts in the former receive significantly more likes (diff. = 0.30, $p < 0.05$) but not more comments (diff. = 0.19, $p > 0.1$), suggesting that the content of comments partially drives the observed positive effects. Moreover, we observe insignificant differences in likes and comments between the bot-comment group and the human-comment group, suggesting that the bot identity alone does not drive enhanced user engagement. In summary, the combination of the comment content and the bot identity is the source of the positive effect of bot comments on user engagement.

Moreover, based on participant responses to open-ended follow-up questions, we conduct a qualitative analysis to further explore the mechanisms driving user engagement, with procedures reported in Online Appendix D. Interestingly, the qualitative analysis provides supporting evidence that relevant bot comments with appropriate social cues can serve as effective social proof, encouraging human users to engage. We report the evidence in Online Appendix D and, in this subsection, focus on the quality attributes of bot comments—relevance and social cues—that justify their role as a good social actor. Table D.2 in the Online Appendix presents summary statistics of responses, whereas Table D.3 provides examples of participant responses to illustrate how users value relevance and social cues in their engagement decisions. First, relevance emerges as a key factor driving user engagement across all post types. For emotional posts, most participants reference relevance (90.7% for likes, 78.4% for comments), emphasizing the importance of bot comments that directly addressed the original post's content. One participant notes (row 1, Online Appendix Table D.3), "Because it is very good at correctly answering the questions and it understands the need." In subjective posts, relevance remains critical, with 92.3% of participants citing relevance for likes and 56.3% for comments. For example, one participant remarks (row 4, Online Appendix Table D.3), "The answer is very thorough, very straight to the point." For objective posts, relevance continues to play a significant role (80.6% for likes, 57.9% for comments). As one participant states (row 7, Online Appendix Table D.3), "CommentRobot accurately and professionally answered the math problem, demonstrating the value of AI in knowledge dissemination."

Second, participants highlight the bot's ability to convey human-like social cues as a key factor in enhancing

**Figure 2.** (Color online) Effects of Bot Comment on Engagement



engagement—particularly its ability to provide emotional support and express subjective opinions. For emotional posts, a significant portion of participants reference social cues (61.1% for likes, 51.4% for comments). One participant notes (row 11, Online Appendix Table D.3), "It comforts people like a real person, making people feel very comfortable." Similarly, for subjective posts, participants mention social cues embedded in bot comments (33.8% for likes, 18.8% for comments). As one participant states (row 14, Online Appendix Table D.3), "It can provide different opinions like a friend." For objective posts, no participants mention social cues, indicating that users do not expect socially expressive responses in factual discussions.

**5.4.2. Effects of the Quality of Bot-Generated Comments.** Inspired by the results of the online experiment, we further leverage the observational data to empirically validate whether the quality of bot comments, as captured by three key constructs (attractiveness, relevance, and social cues), serves as the mechanism driving user engagement. First, Weibo allows users to like comments under a post, providing a natural metric for evaluating how engaging and attractive a comment is to the audience. We construct the variable *BotCommentAttractiveness* to indicate the number of likes received by a bot comment. Second, we employ the GPT-4o-mini model[3] with prompt engineering for few-shot classification to develop a text classifier that assesses the relevance of bot comments to user posts. Based on this classifier, we construct a binary variable, *BotCommentRelevance*, to indicate whether a bot comment is relevant to the post. The details are reported in Online Appendix F.

Social cues reflect how human-like a bot comment is. From our classification of posts into five categories (see Online Appendix E for details), we focus on the categories where social cues are most relevant: subjective questions and emotional support posts. Therefore, we consider the degree of emotional support and subjectivity in bot comments to be measures of social cues. Utilizing the GPT-4o-mini model with prompt engineering for few-shot classification, we build text classifiers to measure these social cues in bot-generated comments. Online Appendix F reports the details. Based on these classifiers, we construct two binary variables—*BotCommentSubjectivity* and *BotCommentEmotional*—indicating whether a bot comment exhibits subjectivity and provides emotional support, respectively.

We impute *BotCommentAttractiveness*, *BotCommentRelevance*, *BotCommentSubjectivity*, and *BotCommentEmotional* to zero for the control group.[4] Table 5 reports the results when replacing *BotComment* with one of these variables in Model (1). The significantly positive coefficients of *BotCommentAttractiveness*, *BotCommentRelevance*, *BotCommentSubjectivity*, and *BotCommentEmotional* indicate that bot comments that are more attractive, relevant, subjective, and emotionally supportive exert stronger effects on user engagement. These results validate our proposed mechanism through which the quality of bot comments amplifies user engagement.

## 5.5. Bot Targeting Strategy

This section is dedicated to scrutinizing the efficiency of the current targeting strategy and exploring strategies to further improve it.

**5.5.1. Efficiencies of Existing Strategy.** First, with *BotComment* as the outcome variable, we investigate how user- and post-level characteristics affect the likelihood of receiving a bot comment. To this end, we estimate linear probability models and logistic regressions, which are reported in Table H.1 in the Online Appendix. We observe

**Table 5.** Effects of the Quality of Bot-Generated Comments

| | *LogLikes* | | | | *LogComments* | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *BotCommentAttractiveness* | 0.0019***<br>(0.0006) | | | | 0.0011***<br>(0.0003) | | | |
| *BotCommentRelevance* | | 0.1303***<br>(0.0070) | | | | 0.2093***<br>(0.0062) | | |
| *BotCommentSubjectivity* | | | 0.2042***<br>(0.0133) | | | | 0.1886***<br>(0.0101) | |
| *BotCommentEmotional* | | | | 0.0697***<br>(0.0136) | | | | 0.1556***<br>(0.0120) |
| User controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Post controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Date FEs | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Hour-of-day FEs | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 64,365 | 64,365 | 64,365 | 64,365 | 64,365 | 64,365 | 64,365 | 64,365 |
| $R^2$ | 0.184 | 0.170 | 0.169 | 0.165 | 0.176 | 0.183 | 0.174 | 0.171 |

*Note.* Robust standard errors are in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

a certain level of efficiency in the strategy. As evidenced by the significantly positive coefficients of *ContentLength* across all specifications, the bot prioritizes longer posts, which may not be inefficient because of the mixed moderating effects of *ContentLength* (see columns (1) and (5) of Online Appendix Table H.2). Meanwhile, posts that do not mention other accounts are prioritized by the bot, as shown by the significantly negative coefficients of *AtOthers* in Table H.1 in the Online Appendix. This strategy is efficient because mentioning others in a post attenuates the positive effects of bot comments on user engagement (see columns (2) and (6) of Online Appendix Table H.2).

However, a higher level of inefficiency is observed. The significantly negative coefficients of *LogUpdates* in Table H.1 in the Online Appendix suggest a preference for commenting on posts from less active users, potentially aiming to boost their activity. Nonetheless, because users who post more frequently benefit more from bot comments than idle users (see columns (3) and (7) of Online Appendix Table H.2), the current targeting practice appears to be inefficient. Meanwhile, we find that posts starting with a mention of the bot are deprioritized, as indicated by the negative coefficients of *AtBotAsStart* in Table H.1 in the Online Appendix. This strategy is not productive because such posts generate a higher engagement level in terms of both likes and comments (see columns (4) and (8) of Online Appendix Table H.2).

Moreover, we fine-tune the GPT-4o-mini model to classify bot-related posts on Weibo into five categories: (1) *Comment Request*, in which users simply request the bot to comment on their posts; (2) *Subjective Question*, in which users request the bot to share views, comments, and suggestions on a topic; (3) *Objective Question*, in which users ask the bot to address an objective question; (4) *Emotional Support*, in which users share their feelings, daily life, and thoughts; and (5) *About Bot*, in which users investigate the bot's abilities, status, etc. (see Online Appendix E for details). The coefficients of categories 2–5 in Table H.1 in the Online Appendix are consistently negative and statistically significant, demonstrating the bot's preference for simple comment requests over other categories. This practice is not in the best interest of the platform and users because, as Table H.3 and Figure H.1 in the Online Appendix reveal, the other four categories receiving bot comments could attract more engagement from the audience.

**5.5.2. Policy Learning.** We rely on the double machine learning approach proposed by Chernozhukov et al. (2018) and the policy tree approach developed by Athey and Wager (2021) to determine the optimal treatment assignment policy based on observational data. First, we estimate an interactive regression model (IRM) that allows treatment effect heterogeneity by modeling interactions between the treatment and control variables. The estimation is conducted using the *DoubleML* Python package developed by Bach et al. (2022). The model controls for all post and user characteristics listed in Table 1, with the sum of likes and comments a post receives as the outcome variable. The estimated average treatment effect of *BotComment* is 6.24 (95% confidence interval (CI): [3.02, 9.46]), indicating a significant positive effect of receiving a bot comment on total engagement. This further reinforces our main findings. Furthermore, the IRM enables us to examine heterogeneous treatment effects across different post and user characteristics. For instance, Figure H.2 in the Online Appendix illustrates the heterogeneity of treatment effects by post category. The observed inefficiencies from our exploratory analysis are reaffirmed by the insignificant treatment effects on posts with comment requests, despite such posts receiving a higher proportion of bot comments (see Online Appendix Table H.1).

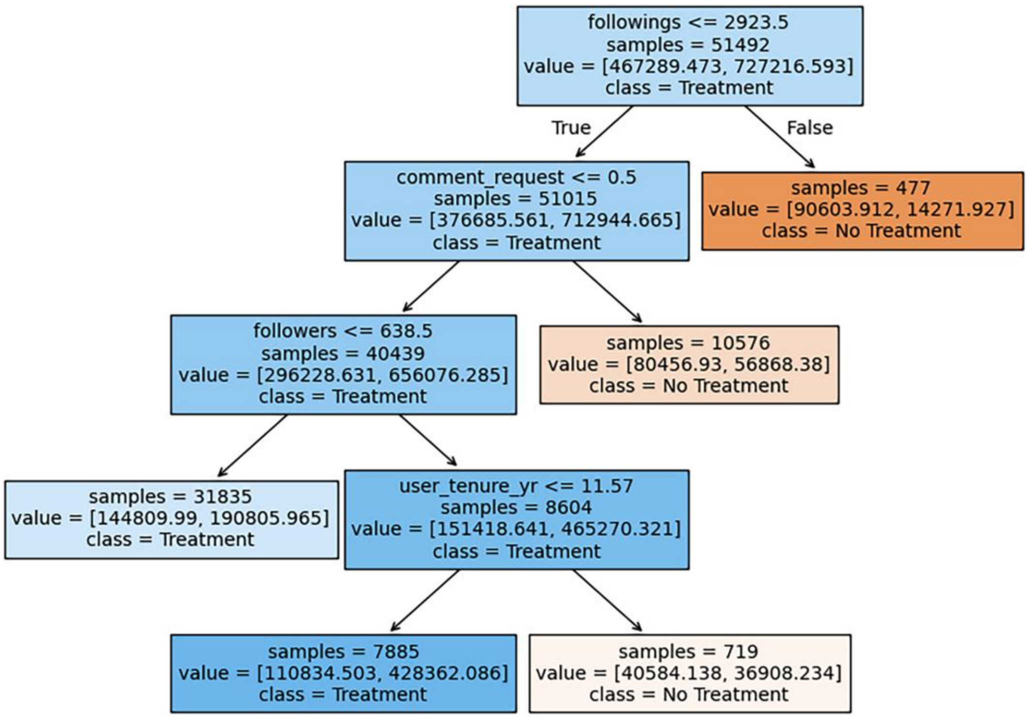
Next, we employ the policy tree approach, a decision-tree-based method designed for optimal treatment assignment in causal inference when treatment effects vary across subgroups (Athey and Wager 2021). Specifically, the policy tree approach seeks to identify the optimal treatment assignment rule, $\hat{\pi}$, by solving the following optimization problem:

$$\hat{\pi} = argmax\left\{\frac{1}{n}\sum_{i=1}^{n}(2\pi(X_i) - 1)\hat{\Gamma_i} : \pi \in \Pi\right\},$$

where $n$ is the sample size, $\pi(X_i)$ is the policy function based on observable features, $X_i$, and $\hat{\Gamma_i}$ represents the doubly robust scores obtained from the above double machine learning approach.

To construct the policy tree, we split the data set into a training set (80%) for policy learning and a test set (20%) for evaluation. One key advantage of the policy tree approach is its flexibility in defining the set of features used for policy learning, allowing us to explore different policy configurations. In the first policy, we assume that the platform aims to leverage all post and user characteristics. Figure 3 presents the resulting policy tree, which suggests a more efficient bot targeting strategy. For example, the policy dictates that users with fewer than 2,924 followings who simply request bot comments should not receive them. This refinement directly addresses the inefficiency noted earlier—bot comments are frequently assigned to posts with comment requests, despite the lack of significant treatment effects on these posts (see Online Appendix Figure H.2). To evaluate the effectiveness of this policy, we follow the methodology outlined by Bach et al. (2022) and compare the average treatment effect on the treated (ATT) under the original treatment policy and the estimated optimal treatment policy in the test set. The estimated ATT under the original policy is 1.99 (95% CI: [−2.73, 6.72]), whereas under the optimized policy, it increases

**Figure 3.** (Color online) Policy Tree Using Post and User Characteristics




to 3.43 (95% CI: [0.86, 6.00]). These results indicate that the optimized policy significantly improves the efficiency of bot targeting strategies.

Moreover, we consider a policy where the platform optimizes bot targeting using only post characteristics, excluding user-level data. The resulting policy tree, displayed in Figure H.3 in the Online Appendix, differs from the previous one. For instance, posts with more than five images are excluded from receiving bot comments. Despite using fewer features, this policy still enhances efficiency, as evidenced by the estimated ATT of 3.54 (95% CI: [1.00, 6.08]). Finally, we examine a policy that relies solely on user characteristics for bot targeting. The policy tree, depicted in Figure H.4 in the Online Appendix, provides another alternative assignment strategy. Although this policy does not leverage post-level features, it still leads to efficiency gains, with an estimated ATT of 2.65 (95% CI: [0.62, 4.67]).[5]

Overall, our findings demonstrate that policy trees can substantially improve bot targeting efficiency by leveraging treatment heterogeneity. The optimized treatment policies consistently yield higher engagement effects compared with the platform's current bot assignment strategy. Additionally, the flexibility of the policy tree approach allows platforms to selectively incorporate features when constructing the tree, balancing optimization with practical considerations such as cost or potential ethical concerns. These results provide actionable insights for platforms aiming to enhance content engagement through AI-driven interventions.

## 6. Does the Bot's Socializing Value Influence User Posting Activity?

Our analyses provide evidence supporting the socializing value of social bots at the post level. In other words, the platform reaches its objective of enhancing user experience. What about the platform's other objective of stimulating user posting activity? To what extent does the socializing value at the post level translate into tangible enhancements in user posting activity across the platform? Answering this question is imperative for platforms seeking to foster thriving online communities.

To this end, we construct a comprehensive user panel data set comprising historical posts made by users in our sample up until January 31, 2024. The outcome variable of interest is users' posting activities, quantified by metrics, including the number of posts mentioning the bot, original posts (excluding those mentioning the bot), and reposts. To ensure a clean identification of treatment effects, we refine our sampling strategy as follows: the treated group consists of users who receive their first bot-generated comment on a specific day—considered the treatment day.[6] For each treatment day, the control group includes users who posted a request on the same day but have never received bot-generated comments within the observation window. Note that users in both treated and control groups could have mentioned the bot more than once.

Conceptually, our data set resembles a series of quasi-experiments, with each day in January 2024 representing a distinct experimental unit. On any given day, a subset of users interacts with the bot, with only a fraction of them receiving bot comments (i.e., the treated group). Leveraging this structure, we employ a stacked DID approach (Cengiz et al. 2019, Zhang et al. 2024), aggregating these quasi-experiments to estimate the average treatment effects. Specifically, users who mention the bot on the same day are grouped into cohorts, facilitating the analysis of treatment effects within these cohorts over time. The estimation window is seven days before and after the treatment day when a user receives their first bot-generated comment. The earliest cohort in our sample is January 9, and the last cohort is January 23. The model specification is as follows:

$$Y_{ist} = \beta_0 + \beta_1 D_{ist} + \lambda_{is} + \delta_{st} + \epsilon_{i,s,t}, \qquad (2)$$

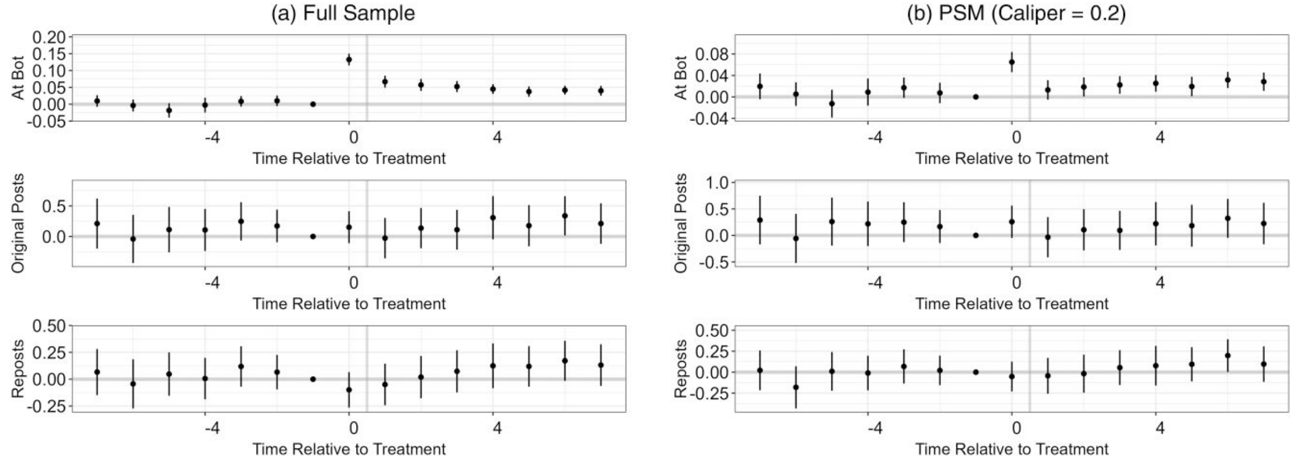
where $Y_{ist}$ is the number of posts mentioning the bot, original posts, or reposts by user $i$ in cohort $s$ at day $t$. $D_{ist}$ indicates whether user $i$ in cohort $s$ at day $t$ has received a bot comment. $\beta_1$ is the coefficient of interest. $\lambda_{is}$ and $\delta_{st}$ denote user-cohort fixed effects and time-cohort fixed effects, respectively. We cluster the standard errors at the user-cohort level.

Figure 4(a) illustrates the results of the stacked DID event study, providing insights into the impact of bot comments on user activity. During the pretreatment periods, we find insignificant differences in all three outcome variables between the treated and control groups, supporting the validity of the parallel trend assumption. Interestingly, in the posttreatment periods, we observe a significant effect of receiving bot comments on the number of posts mentioning the bot, indicating that focal users enjoy the engagement lifted by the bot comment and are thus willing to engage more with the bot. Therefore, this finding connects the post-level analysis to the user-level analysis: enhanced user engagement at the post level leads to more bot-related posts. However, we do not observe a significant increase in the number of original posts and reposts during the posttreatment periods.

Nonetheless, selection bias due to the bot's potential targeting strategy remains a major concern in the DID analysis. A common approach in the literature to alleviate the selection concern is propensity score matching (PSM), which further balances the treated and control groups to make them more comparable in terms of observables. Specifically, in each cohort, we match treated users with control users based on their activities in December 2023 and online profiles. The results of the stacked DID event study with PSM are presented in Figure 4(b), consistent with those from Figure 4(a). We try different specifications (e.g., different calipers) of PSM and alternative matching methods (e.g., CEM), all of which yield consistent results.

In conclusion, these results suggest that the socializing value delivered at the post level does boost users' posting frequency, but this increase is only confined to posts seeking bot comments. A potential explanation for this finding is the distinction between engagement and content creation behaviors on social media. Although bot-generated comments can stimulate interaction within a given post, they do not necessarily enhance a user's intrinsic motivation to create additional, unrelated content. According to Oh and Syn (2015), the top five motivations for X/Weibo users are learning, social engagement, self-efficacy, reputation, and reciprocity. However, the current bot design primarily triggers social engagement, without effectively addressing the other motivations that drive broader content generation. As a result, users may perceive bot interactions as event-driven engagement, where their peers respond to the stimulus (bot-generated comments), but they do not see it as a compelling reason to modify their broader posting behavior. This suggests that bot-driven

**Figure 4.** Stacked DID Event Study




*Note.* Relative period −1 serves as the baseline period.

engagement primarily reinforces a *reactive* participation model, where users interact in response to a trigger (bot-generated comments), rather than an *active* content-generation cycle, where users proactively initiate new discussions. In this sense, our findings suggest a potential boundary condition for social bots: their effectiveness in stimulating broader user activities depends on whether they function as reactive or proactive agents in user engagement.

## 7. Conclusion

Leveraging a platform-owned social bot on Weibo, this study delves into the socializing value of LLM-based social bots in the context of public human-bot interactions on social media. A series of analyses provides empirical evidence suggesting that bot-generated comments help posts attract more likes and comments from the audience, thereby validating the socializing value for the focal user at the post level. Mechanism tests further reveal that the socializing value hinges on the bot-generated content, demonstrating that higher-quality (i.e., more attractive, relevant, subjective, and emotionally supportive) comments yield greater increases in user engagement. Moreover, we utilize heterogeneity tests and a policy learning approach to highlight the importance of considering contextual factors to optimize the bot's socializing value. Finally, our investigation into the broader impact of bot interactions on user posting activity reveals that the socializing value delivered at the post level does boost users' posting frequency, but this increase is only confined to bot-related posts.

### 7.1. Theoretical Contribution

This study makes important contributions to the literature. First, this study contributes to the literature on conversational AI and social bots (see Online Appendix Table A.1) by investigating an important and timely question of whether social bots have socializing value through public human-bot interactions on social media. To the best of our knowledge, the socializing value of social bots in public settings has been largely unexplored. Previous literature on social bots has predominantly focused on their instrumental roles in information dissemination (e.g., Shao et al. 2018), content moderation (e.g., He et al. 2024), customer service (e.g., Xu et al. 2017), and marketing tasks (e.g., Faggella 2018), with a few studies examining the socializing value in private settings (Ta et al. 2020). Our study fills this research gap and contributes to a deeper understanding of the role of LLM-powered bots in shaping user interaction on social media platforms. Interestingly, the results of the online experiment indicate that the combination of the bot identity and the LLM-generated content creates the bot's positive socializing value. This finding brings new insights to the literature on bot identity effects because previous findings show that disclosing bot identity might affect engagement either negatively or positively depending on the setting (Mozafari et al. 2021). Moreover, by uncovering the heterogeneous effects of bot comments across different post- and user-level characteristics, we advance knowledge on the nuanced dynamics of human-bot interactions in online environments.

Second, this study contributes to the literature on the CASA framework (Gambino et al. 2020) by highlighting the critical role of bot-generated content in human-bot interactions. Our exploration of bot comment characteristics reveals that comments with higher relevance and richer social cues have stronger positive effects on peer user engagement, which aligns closely with the CASA framework highlighting bots' functional competence and social intelligence. Notably, in terms of social cues, our analysis of the attributes of initial human posts indicates that subjective posts and those seeking emotional support garner greater engagement from peer users when the bot responds to them, compared with posts merely requesting bot comments. More importantly, when bot comments are more subjective and more emotionally supportive, their positive effects on user engagement are stronger. To the best of our knowledge, whereas emotional support has been explored in private bot interactions (Ta et al. 2020), subjectivity has not been well studied, especially in the IS literature (see Online Appendix Table A.1). The new affordance of social media platforms (i.e., public nature) allows us to empirically examine a new construct, the attractiveness of bot-generated content to the audience. The results suggest that more attractive bot comments, as evidenced by a larger number of likes received, lead to increased user engagement. This finding extends the CASA framework to encompass the unique dynamics of public human-bot interactions, suggesting that engagement metrics of bot responses can significantly influence the social dynamics on these platforms.

### 7.2. Practical Contribution

This paper provides important insights for social media platforms. First, our findings highlight the socializing value of social bots at the post level, demonstrating their potential to enhance user experience by increasing engagement with the focal author's posts. For platforms seeking to improve user engagement, deploying LLM-based social bots emerges as a promising strategy. However, to maximize their effectiveness, platforms should actively regulate social bots to behave as good social actors. Specifically, our findings suggest that bot comment quality—particularly relevance and social cues—plays a crucial role in driving user engagement. This implies that bot-generated content should not only be contextually aligned with the user's post but also incorporate human-like social cues to maximize user engagement.

Second, our research underscores the necessity for continuous monitoring and optimization of bot deployment strategies to capitalize on their socializing value while mitigating potential negative repercussions. Notably, our findings reveal inefficiencies in the current targeting strategy employed by Weibo's bot across various dimensions. The results of our policy learning analysis reveal that platforms can address these shortcomings by incorporating factors such as post- and user-level characteristics into their bot deployment strategies. As discussed, one key advantage of the policy learning approach is its flexibility in defining the set of features, allowing managers to explore different policy configurations. By tailoring bot interventions to better align with the diverse needs and preferences of users, platforms can cultivate a more engaging and inclusive online environment. Ultimately, this approach holds the potential to foster stronger user engagement and facilitate more meaningful interactions within online communities.

Third, our findings reveal a critical limitation of the current implementation of social bots. Although public human-bot interactions stimulate bot-related discussions, they do not lead to a general increase in user posting activity. As discussed, this finding could be the result of the bot serving as a reactive agent responding to existing content rather than a proactive agent encouraging new content creation. Therefore, to extend social bots' value beyond driving reactive engagement toward encouraging active participation, platforms may need to redefine the role of social bots. One approach could involve designing bot interactions that prompt human users to contribute rather than simply having the bot react to user-generated content. Additionally, integrating AI-assisted content creation tools (Zhang et al. 2024) or personalized engagement nudges could help transition users from passive interaction to more active posting behaviors by affecting their motivations for learning, reputation, and broader social engagement.

For social media users, our findings suggest that the social bot holds the potential of acting as a positive social actor. Previous literature suggests that social media engagement metrics are important to the focal author. For example, receiving more likes and comments has been associated with increased self-esteem and happiness, and users who receive more comments tend to perceive the social media platform as more interesting and caring (Zell and Moeller 2018). Additionally, the psychological need for social rewards, such as likes, plays a significant role in user behavior (Lindström et al. 2021). Given that bot-generated comments significantly increase engagement at the post level, they may positively influence the focal author's perception of the platform and overall well-being. Moreover, social bots can provide emotional support, potentially benefiting users who seek affirmation or empathy in online interactions. Our sentiment analysis comparing direct replies to bot comments with user-only interactions finds no significant differences in sentiment (see Online Appendix Figure G.3), suggesting that users do not perceive bot interactions as riskier or less meaningful than human interactions. This reinforces the generally positive perception of social bots within the social media community, suggesting that users may welcome their presence as a source of engagement and support.

For policymakers, the results of the online experiment suggest that revealing the bot's identity does not diminish its positive effects on user engagement. This finding reinforces the importance of transparency measures, such as clearly labeling bot-generated interactions, to ensure that users are fully aware of AI involvement in social media conversations while maintaining engagement benefits. Transparent disclosure can help build user trust and mitigate concerns about AI-driven manipulation or misinformation.

### 7.3. Limitations and Future Research

Although our study provides valuable insights, it has certain limitations and presents opportunities for future research. First, our exploration focuses on bot interactions in the form of comments, leaving room for future research to examine the socializing value of bots in other modalities, such as likes, shares, and following users. Bots that like or share posts may act as implicit endorsements or social proof, potentially influencing user perceptions and engagement differently than direct comments. Future research could explore whether these alternative interactions foster more sustained engagement or encourage broader content creation.

Second, demographic differences in human-bot interactions remain underexplored. User characteristics such as age, gender, experience with AI, and cultural background may shape how individuals perceive and engage with social bots. Users who are more familiar with AI may respond differently compared with those less experienced with social bots. Future studies could examine whether certain user groups are more likely to benefit from or be influenced by bot-generated interactions.

Third, our study is based on a single platform, which has specific social and technological affordances that shape user interactions. Although our findings provide strong empirical evidence in this context, comparisons across different platforms are needed to assess the generalizability of the results. Social media platforms vary in user norms and interaction structures, which may moderate the effects observed in this study. Future research could explore how platform-specific factors influence the effectiveness of social bots and whether bot-driven engagement differs across various digital environments.

Fourth, our measure of likes is limited by data availability. Unlike comments, which allow us to exclude those posted before a bot comment, the timing of individual likes is not publicly observable, making it impossible to determine whether a like occurred before or after a bot comment. Although the online experiment helps alleviate reverse causality concerns—because participants reported their willingness to like or comment only after seeing a bot comment—we acknowledge that the likes measure may still contain some noise. Future studies with access to timestamped engagement data could more precisely assess how bot-generated comments influence users' liking behavior.

Finally, we note that bot-generated comments fall short in encouraging broader content creation by the focal user. This highlights a potential boundary condition of social bots—their current design supports reactive engagement but may not foster active participation. Future research could explore how bots might be redesigned to serve as proactive agents that not only elicit responses but also motivate users to initiate new posts or discussions.

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## Endnotes

[1] Reasons for user fatigue and disengagement on social media include concerns about the negative impact on mental health, growing privacy issues, tighter regulations, data ownership, and freedom of speech. Considering that social media revenues, often driven by advertisements, depend heavily on active user engagement, addressing fatigue and disengagement is crucial for all platforms.

[2] How a user's commenting behaviors are displayed on Weibo is different from X, another popular microblogging platform. On Weibo, if a user leaves a comment on a post from other users, the comment will not appear in the commenter's followers' feeds nor in the commenter's timeline. However, on X, the comment could appear in the followers' feeds and the commenter's timeline.

[3] For details, see https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/.

[4] We thank an anonymous reviewer for this important suggestion.

[5] User characteristics in our sample may introduce noise or heterogeneity that reduces their effectiveness for policy targeting, which may explain why the highest ATT is achieved when the policy relies solely on post characteristics.

[6] For those treated multiple times, only the day of their first bot comment is considered the treatment day. After the treatment day, users in the treated group remain treated. To address the potential impact of multiple treatments (i.e., receiving multiple bot comments at different times), we conducted an analysis with users who received a bot comment only once as the treated group. The results are qualitatively the same and are available upon request.

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